COMP0005 – ALGORITHMS

Introduction

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Dept of Computer Science

CONTENT

- What the module is about
  - Content
  - Resources
  - Assessment

- How the module will be delivered
  - Asynchronous activities
  - Synchronous activities
  - Communication
  - Teaching team

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COMP0005 – THE “WHAT”

- What is this course about?
  - Algorithm: method to solve a problem
  - (& Data Structure: method to store information)

- Why study them?

COMP0005 – THE “WHAT”

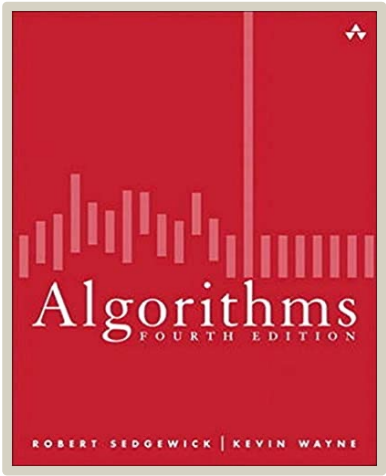
- Five Main Topics:
  1. Fundamentals
  2. Sort Algorithms
  3. Search Algorithms
  4. Graph Algorithms
  5. String Algorithms

COMP0005 – THE “WHAT”

- Five Main Topics:
  1. Fundamentals
     a. Steps to develop an algorithm
     b. Analysis of algorithms
     c. Abstract data types
  2. Sort Algorithms
  3. Search Algorithms
  4. Graph Algorithms
  5. String Algorithms

COMP0005 – THE "WHAT"

- Learning Resources (external)
  - Textbook: Algorithms, by R. Sedgewick & K. Wayne. Addison-Wesley 2011. ISBN: 978-0-321-57351-3
    - Book website: https://algs4.cs.princeton.edu/home/
  - Programming language for practicals: Python 3 & Jupyter Notebook
    - Jupyter website: https://jupyter.org/try



COMP0005 – THE "WHAT"

- Assessment
  - Group coursework: 20%
    - Released March 6th, 2025; due March 24th, 2025
      - Marks & feedback returned by April 25th, 2025
    - Group vs individual marking
  - Individual 2-hour on-campus (invigilated) exam: 80%
    - Term 3 (date TBA)

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  - Asynchronous activities
  - Synchronous activities
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COMP0005 – THE “HOW”

- Asynchronous activities
  - Pre-recorded Lectures
  - Self-practice Exercises

COMP0005 – THE “HOW”

- Synchronous activities
  - On-campus Lectures (1h/week)
  - On-campus Labs with Teaching Assistants (2h/week)
  - On-campus Office Hours (1h/week)

COMP0005 – THE “HOW”

- Synchronous activities
  - On-campus Lectures (1h/week)
  - On-campus Labs with Teaching Assistants (2h/week)
    - Wednesdays 9am-11am
    - Three sessions in parallel (check in Moodle where to go)
    - Work within your COMP0005 coursework team
  - On campus Office Hours (1h/week)

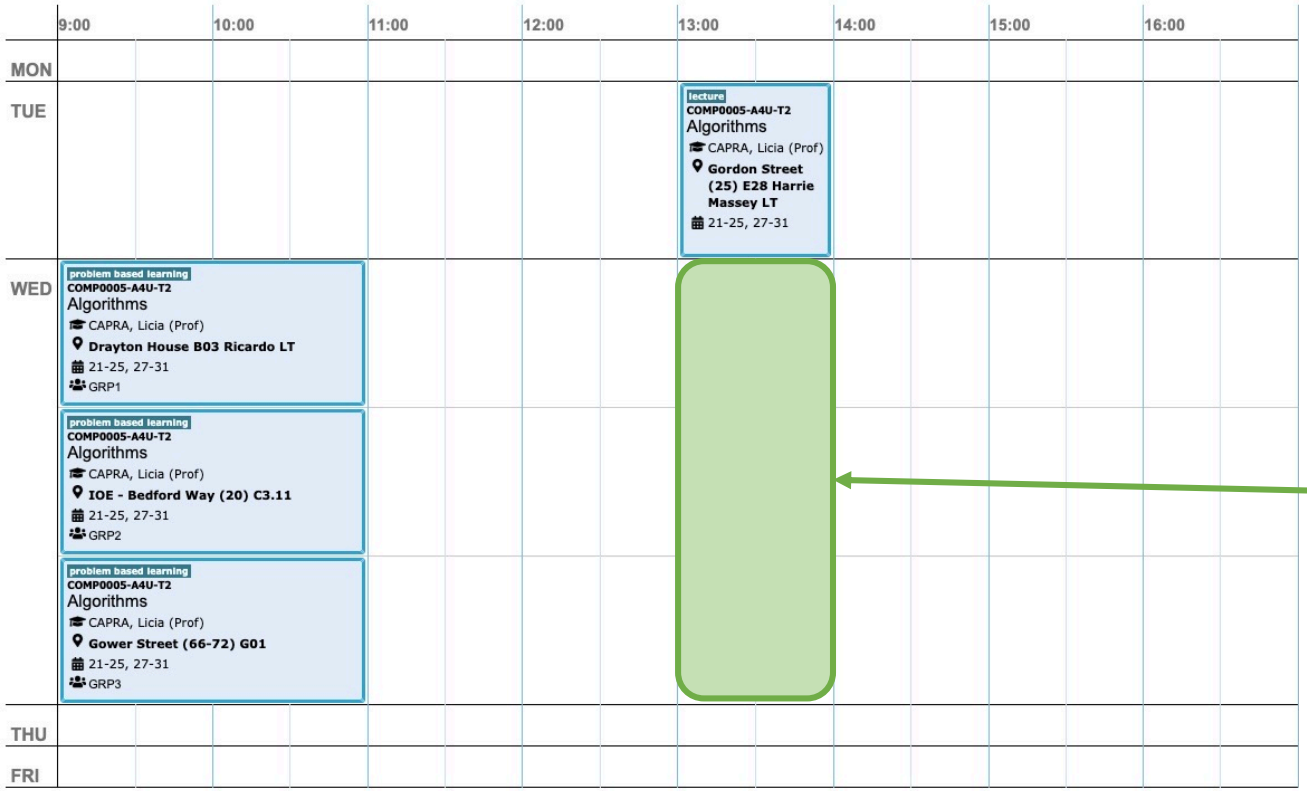
COMP0005 – THE “HOW”

- Synchronous activities
  - On-campus Lectures (1h/week)
    - Tuesdays 1pm-2pm, Harrie Massey LT
  - On-campus Labs with Teaching Assistants (2h/week)
  - On-campus Office Hours (1h/week)

COMP0005 – THE “HOW”

- Synchronous activities
  - On-campus Lectures (1h/week)
  - On-campus Labs with Teaching Assistants (2h/week)
  - On-campus Office Hours (1h/week)
    - Wednesdays, 1pm-2pm, room 508 @ 66-72 Gower Street
    - Optional, drop in

COMP0005 – THE “HOW”



Optional drop-in hour for Q&A
(508 @ 66-72 Gower St)

COMP0005 – THE “HOW”

- Communication
  - Moodle Forums (News, Q&As)
  - Moodle (fortnightly) “Muddiest Point” (closes on Weds at 23:55)
  - Office Hours
  - Email

  - *DO NOT use MS Teams Chat*

COMP0005 – THE “HOW”

- Teaching Team
  - Lecturer: Licia Capra
  - Teaching Assistants
    - Carol & Kyle: GRP1 (Drayton House B03 Ricardo LT)
      - COMP0005 groups 1 to 16
    - George & Erin: GRP2 (IOE – Bedford Way (20) C3.11)
      - COMP0005 groups 17 to 31
    - Nakul & Shreevanth: GRP3 (G01 66-72 Gower Street)
      - COMP0005 groups 32 to 42
  - Administrators: cs.undergraduate-students@ucl.ac.uk

COMP0005 – THE “HOW”

- Moodle module website

COMP0005 – THE "HOW"

- A typical week (~10-15 hours of work)
  - Lectures (3h per week)
    - Watch pre-recorded lectures
    - Tuesday synch session

    Mondays & Tuesdays
  - Independent study (~6h+ per week)
    - Study of lecture material
    - Work on Self-Practice exercise sheets
    - Work on TA lab sheets

    Mondays & Tuesdays
  - Lab (2h per week)

    Wednesdays

RECAP

- *What the module is about*
- *How the module will be delivered*

- What next
  - Today/ASAP:
    - watch pre-rec lectures (& slides) in Moodle under Week 1
    - work on 1st self-practice exercise sheet
  - Tomorrow (Wednesday 9am-11am):
    - join your first TA lab, meet your TAs, meet your COMP0005 group
    - work on 1st TA lab sheet
  - See you next Tuesday @ 1pm